



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2019/2020

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
OF SCIENCE IN ELECTRICAL AND ELECTRONICS ENGINEERING
EEE 205: PHYSICAL ELECTRONICS

DATE: Wednesday 5th February 2020

TIME: 11.00a.m. - 1.00p.m.

INSTRUCTIONS

Question ONE is compulsory

Answer any other TWO questions

Planks constant = 6.626×10^{-34} J-sec.

Electronic charge = 1.6×10^{-19} C

1eV = 1.6×10^{-19} Joules

Mass of electron, $m_e = 9.1 \times 10^{-31}$ kg

Permittivity of free space, $\epsilon_0 = 8.854 \times 10^{-12}$ F-m

$K = 1.38 \times 10^{-23}$



Question 1

a. With aid of a VI- characteristic, briefly describe the behavior of a tunnel diode when reverse and forward biased. (5 marks)

b. ~~With aid of a labelled diagram of voltage against current characteristic, describe the three electric discharge regimes namely dark discharge, glow discharge and arc discharge.~~ (5 marks)

c. A certain atom has 12 electrons in total. For this atom, determine:

- Total energy possessed by electrons in 1st and 2nd orbits
- Frequency of electromagnetic radiation emitted when electron falls from 2nd to 1st orbit.
- Wavelength of the emitted wave.
- Radius of 1st orbit

d. A germanium semiconductor is doped with 3×10^{16} atoms of phosphorus per cm^3 and 6×10^{16} boron atoms per cm^3 . Determine:

- Type of the impure semiconductor.
- Majority and minority charge carriers.

(4 marks)

 A light emitting diode with minimum and maximum voltage drops of 1.8 volts and 3 volts respectively is connected alternately with 24 volt supply (with 820 ohm resistor) and 5 volts supply (with 120 ohm resistor). Indicate which supply voltage will keep the brightness of the diode constant.

(6 marks)

Question 2

a. Describe what is meant by mean free path.

(2 marks)

b. The expression of Schrodinger time independent wave equation is given by $E\Psi = \frac{\hbar^2}{2m} \frac{d^2\Psi(x)}{dx^2} + V\Psi(x)$. Use the expression to derive Schrodinger time dependent wave equation.

(5 marks)

c. Calculate the voltage which has to be applied across a PN semiconductor junction to cause a forward current of 5 A/ cm^2 , assuming that the saturation current density at 300K is 25 micro ampere per cm^2 .

(5 marks)

d. Briefly describe the four quantum numbers.

(8 mark)

Question 3

a. Derive the expression of mobility of electrons in a good conductor.

(3 marks)

b. Briefly describe the two parameters of intrinsic semiconductors that determines its

conductivity, and describe how they vary with changes in temperature.

(4 marks)

c. State any THREE Bohr atom postulates.

(3 marks)

d. A semiconductor at 300K has intrinsic concentration, $n_i = 1.5 \times 10^{16}/\text{m}^3$ and resistivity of $\rho = 2 \times 10^{13}$ ohm-meter. Donor impurity atoms $N_D = 1 \times 10^{20}$ per m^3 were added as dopants.

Determine the shift in Fermi energy level due to doping in electron-volts.

(5 marks)

e. Find the velocity of charge leading to a current of 2 A which flows in an aluminium conductor of cross section 2 cm^2 and length 15 kilometers. Free electron density of aluminium is $12 \times 10^{28}/\text{m}^3$. How long will it take for the electric charge to travel from one end of the conductor to the other.

(5 marks)

- Describe the mass action law and give its expression. (3 marks)
- b. Describe any three applications of light emitting diode. (3 marks)
- c. An earth satellite has on board 12V battery which supplies a continuous current of 1A. Solar cells are used to keep the battery charged. The solar cells are illuminated by sun for 12 hours in every 24 hours. If the during exposure, each cell gives 0.5V at 100mA, determine the number of cells required. To charge the battery 13.0 volts are needed. (6 marks)
- d. The Fermi energy level of a particular semiconductor material is 6.25eV at $T=300K$. assuming the electrons in the material follow Fermi Dirac distribution function, determine the following:
- The probability of energy level at 6.25eV being occupied by an electron.
 - Repeat part (i) if the temperature is increased to $T=950K$.
 - The temperature where the probability that a state of 0.30eV below the Fermi level is 0.01. (8 marks)

Question 5

- a. Derive the expression to proof that the Fermi level is mid-way between conduction band and the valence band in a pure semiconductor. (5 marks)
- b. A pure germanium semiconductor has 8×10^{30} atoms per m^3 at 300K. There are 2×10^{16} electron hole pairs at 300K. Assume mobility of electrons is $0.39m^2/v-s$ and holes $0.19m^2/v-s$
- Find resistivity of germanium at this temperature.
 - The conductivity of germanium increases at the rate of 6% per degree centigrade. Find the resistivity of germanium at 400°K.
 - If the pure germanium is doped with acceptor atoms at the rate of 10 atoms per 10^8 of germanium atoms, find the resistivity of the doped semiconductor at 300°K. (10 marks)
- c. Determine the applied voltage to achieve a forward current of 0.5 μA in a PN junction silicon diode at 27°C, if the reverse saturation current is 0.8 nA. If the reverse saturation current in a germanium diode is 8 μA , what current would result if the voltage determined is applied in the forward direction assuming $\eta=1$ for germanium and 2 for silicon diodes. (5 marks)

$T = 300K$
 $I_s = 0.8 \times 10^{-9}$

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FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE ELECTRICAL AND ELECTRONIC ENGINEERING AND BIOMEDICAL
ENGINEERING

EEE 205: PHYSICAL ELECTRONICS

DATE: Tuesday, 16th December, 2014

TIME: 2.00 p.m. – 4.00 p.m.

INSTRUCTIONS: ATTEMPT QUESTION ONE AND ANY OTHER TWO QUESTIONS

Use: $e = 1.602 \times 10^{-19} \text{ C}$, $h = 6.626 \times 10^{-34} \text{ J-s}$, $m_e = 9.109 \times 10^{-31} \text{ kg}$, $K = 1.381 \times 10^{-23} \text{ J/K}$, $N_A = 6.023 \times 10^{23} \text{ molecules/mole}$, $\text{eV} = 1.602 \times 10^{-19} \text{ J}$, $c = 2.998 \times 10^8 \text{ m/s}$

Question one (30marks)

- (a) Distinguish between intrinsic and extrinsic semiconductors (2marks)
- (b) Explain the terms:
- (i) Fermi level – energy level (2marks)
 - (ii) work-function (4marks)
- (c) Explain the following
- (i) Rubber is an insulator while copper is a conductor (2marks)
 - (ii) In intrinsic semiconductors the conductivity is low (2marks)
 - (iii) Addition of small amounts of impurities increase the conductivity of semiconducting materials (2marks)
 - (iv) Silicon is preferred as compared to germanium as a semiconducting material

(d) Explain Fermi Dirac Distribution function. Draw a graph showing this function (5marks)

(e) Draw and explain the V-I characteristics of a semiconductor diode - practical (5marks)

(f) With the aid of a diagram, discuss the working of (i) Zener diode (ii) varactor diode (6marks)

(g) State the Pauli Exclusion Principle (2mark)

Question two (20marks)

(a) With the aid of diagrams, discuss the formation of energy bands in solids and explain how they differ in conductors, insulators and semiconductors (9marks)

(b) Find the resistivity of intrinsic silicon at 300k. Assume $n_i = 1.5 \times 10^{10}/\text{cm}^3$, $\mu_n = 1300 \text{ cm}^2/\text{V-sec}$ and $\mu_p = 500 \text{ cm}^2/\text{V-sec}$ (4marks)

(c) (i) Write down the assumptions of classical free electron theory (4 Marks)

(ii) Explain failure of classical free electron theory (3 Marks)

Question three (20marks)

(a) Show that Fermi level in an intrinsic semiconductor lies in the Centre of forbidden energy gap. Explain the effect of doping on the Fermi level. Draw diagrams to illustrate your answer (6marks)

(b) A conducting line on an IC chip has a length of 2.6mm and area of cross-section of $(1.2 \times 10^{-6}) \times (4 \times 10^{-6})$ mm. A current of 4.5mA causes a voltage of 110mV across the line. Find the electron concentration if electron mobility is 500m²/V-s. (3marks)

(c) A silicon diode is operating at 300k. For a forward voltage of 0.7V the forward current is 1×10^{-3} A.

(i) If $n = 2$ find the current if voltage is 0.08V (3marks)

(ii) Determine reverse saturation current. (3marks)

(d) Explain how the Bohr model and the quantum mechanical model of the atom differ in how they describe electrons - maxwell, fermi dirac - any other, quantum. (5marks)

Question four (20marks)

STUDENT CENTRE K.M
Email: studentcentre@yahoo.com

- (a) Describe how quantum free electron theory has been successful in overcoming the failures of classical free electron theory (6 Marks)
- (b) Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K, in a material. (4 Marks)
- (c) A silicon p-n junction has a reverse saturation current of $30\mu\text{A}$ at 25°C . Find static and dynamic resistance at a forward voltage of 0.3V. (4marks)
- (d) Define drift velocity, mobility and relaxation time for free electron and derive the expression for conductivity in metals (6marks)

Question five (20marks)

- (a) Describe the following in relation to quantum mechanical model and indicate their possible values (i) principal quantum number (2marks)
- (ii) Angular momentum quantum number — (2marks)
- (iii) Magnetic quantum number (2marks)
- (iv) Electron spin quantum number (2marks)
- (b) State and explain Schrodinger equation (3marks)
- (c) Differentiate between arc and glow discharge and give three applications of each (9marks)



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN ELECTRICAL AND ELECTRONICS ENGINEERING

EEE 205: PHYSICAL ELECTRONICS

DATE: TUESDAY 27TH FEBRUARY 2018

TIME: 11.00 A.M. - 1.00 P.M.

INSTRUCTIONS

Attempt Question ONE and any other TWO

QUESTION ONE

- a) Explain the meaning of the following terms as used in semiconductor physics;
- Fermi-Level
 - Valence energy band
 - Electron-hole recombination
- (6 marks)
- b) State the Pauli Exclusion Principle and explain its relevance in quantum mechanics
- (3 marks)
- c) Distinguish between intrinsic and extrinsic semiconductors Give an example of each.
- (4 marks)
- d) Sketch a suitable current-voltage (I-V) diagram of a PN Junction and explain the breakdown mechanism
- (5 marks)
- e) Mention three ways of increasing the efficiency of Photo-Voltaic cells
- (3 marks)
- f) Consider an abrupt PN junction in reverse bias with depletion layer thickness, W . Use suitable derivations to show that the maximum electric field, $E_{\max}$ generated across the PN junction at zero-bias condition is given by;

$$E_{\max} = \frac{qN_D x_n}{\epsilon_s}$$

Where q , N_D , ϵ_s are the carrier charge density, doping element concentration and permittivity of the semiconductor material respectively

(8 marks)

QUESTION TWO

- a) Using energy band diagrams, explain difference between conductors, semiconductors and insulators (5 marks)
- (b) What are the main three methods of moderating the concentration of elections and holes in a semiconductor material? (6 marks)
- c) An atom X is doped with two dopants of different concentrations; $1 \times 10^{16} \text{ cm}^{-3}$ of Phosphorus (P) and $2 \times 10^{16} \text{ cm}^{-3}$ of Boron (B). Determine the type of material formed after doping, and the electron and hole carrier concentrations (3 marks)
- d) The Fermi energy level of a semiconductor material is 4.25 eV at room temperature. Assuming that the electrons in the material follow a Fermi-Dirac distribution function, determine the probability of an energy level at 3.25 eV being occupied by electrons. What is the temperature at which the probability of a state of 0.30 eV below the Fermi level being empty is 1%? (6 marks)

QUESTION THREE

- a) Use suitable diagrams to explain the operation of a PN junction in reverse bias (4 marks)
- b) By using suitable diagrams, explain the principle of tunneling as applied in Varactor diodes (5 marks)
- c) Use a suitable diagram to describe the I-V characteristics of a Zener diode and clearly show its operating region (4 marks)
- d) An abrupt silicon PN junction is formed by creating an implant of $N_A = 5 \times 10^{16} \text{ holes / cm}^3$ into an *n-type* substrate of doping $N_D = 10^{15} \text{ electrons / cm}^3$. Determine;
- i. Built-in potential of the junction, ϕ_o
 - ii. Thickness of the depletion region, W
 - iii. Maximum electric field, E_{max} generated across the PN junction at zero-bias condition Assume $T = 300 \text{ K}$ (7 marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2015/2016

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN ELECTRICAL AND ELECTRONICS ENGINEERING

EEE 205: PHYSICAL ELECTRONICS

DATE: FRIDAY 11TH DECEMBER 2015

TIME: 11.00 A.M. - 1.00 P.M.

INSTRUCTIONS

This paper contains FIVE Questions

Attempt Question ONE and any other TWO

Use $e = 1.602 \times 10^{-19} \text{ C}$, $k = 1.381 \times 10^{-23} \text{ J/K}$, $eV = 1.602 \times 10^{-19} \text{ J}$, $c = 3.0 \times 10^8 \text{ m/s}$, $h = 6.626 \times 10^{-34} \text{ J-s}$

QUESTION ONE

- a) Explain the meaning of the following terms as used in semiconductor devices;
- Electron-hole generation
 - Intrinsic semiconductor
 - Fermi Level
- (3 marks)
- b) Distinguish between majority and minority carriers in semiconductor materials
- (4 marks)
- c) Draw the electrical equivalent of a solar cell and briefly explain all the parameters in the equivalent circuit. Write an expression for the output current of the cell
- (5 Marks)
- d) Use suitable diagrams to explain the transition of electrons between the valence and conduction bands of a semiconductor
- (5 marks)
- e) Sketch and explain the I-V curve of a semiconductor diode
- (4 marks)
- f) Use suitable diagrams to explain the formation of a potential/ junction barrier in P-N junction and show the polarity of the barrier potential
- (4 marks)

- g) Consider an n-type silicon semiconductor at $T=300\text{K}$ in which $N_d=10^{16}\text{cm}^{-3}$. Assuming an intrinsic carrier concentration $n_i=1.5 \times 10^{10}\text{cm}^{-3}$, determine the equilibrium electron and hole concentrations for the given doping concentrations (5 marks)

QUESTION TWO

- a) Write down the time independent form of Schrodinger's equation and explain the meaning of each term (6 marks)
- b) Use suitable diagrams to discuss the formation of energy bands in solids and explain how they differ in conductors, semiconductors and insulators (6 marks)
- c) A Silicon semiconductor has the following parameters at $T=300\text{K}$; electron concentration, $n=10^{17}\text{cm}^{-3}$; hole concentration, $p=10^3\text{cm}^{-3}$; intrinsic carrier concentration, $n_i=10^{10}\text{cm}^{-3}$. Find the Fermi level (with respect to either E_c or E_v) for this semiconductor at room temperature, and draw the energy band diagram. Make sure that you clearly label E_f , E_c , E_v . Boltzmann's constant, $k=8.617 \times 10^{-5}\text{eV/K}$ or $1.38 \times 10^{-23}\text{J/K}$ (8 marks)

QUESTION THREE

- a) A silicon atom is doped with $3 \times 10^{16}\text{cm}^{-3}$ of Phosphorus (P) and $6 \times 10^{16}\text{cm}^{-3}$ of Boron (B) atom. Determine the following;
- Type of material
 - Electron and hole concentrations
 - Majority and minority charge carriers
- b) Consider a PN junction with no voltage applied on its terminals. Draw a band diagram indicating the valence band, conduction band, Fermi energy, built-in potential and the depletion region (6 marks)

(iii) Hence find the reflection and transmission coefficients and show that $R + T = 1$.

What does the form of transmission coefficient suggest? (5 marks)

(iv) Draw a figure showing the variation of probability density $|P(x)|^2$ with respect to x and then explain the tunnel effect. (5 marks)

5.(a) Solve the Schrodinger equation for a particle which is constrained to move inside the one dimensional infinite square well potential (i.e. particle moving in a one dimensional box) to find the eigenfunctions and eigenvalues of the particle.
(The potential is defined as

$$V(x) = \begin{cases} 0, & -a < x < a \\ \infty, & |x| > a \end{cases} \quad (7 \text{ marks})$$

(b) Normalize the wavefunctions thus obtained and show that they are orthogonal. (5 marks)

(c) The energy obtained in (a) is quantized (discrete energy levels). How is this result different from classical mechanics result and why do you get this quantization? (4 marks)

(d) Calculate in eV the energy of the first excited state of a particle of mass 10^{-30} kg confined to one dimensional infinite square well potential of dimension 10^{-14} m . (4 marks)

(iii) Hence find the reflection and transmission coefficients and show that $R + T = 1$.

What does the form of transmission coefficient suggest? (5 marks)

(iv) Draw a figure showing the variation of probability density $|\Psi(x)|^2$ with respect to x and then explain the tunnel effect. (5 marks)

5.(a) Solve the Schrodinger equation for a particle which is constrained to move inside the one dimensional infinite square well potential (i.e. particle moving in a one dimensional box) to find the eigenfunctions and eigenvalues of the particle.
(The potential is defined as

$$V(x) = \begin{cases} 0, & -a < x < a \\ \infty, & |x| > a \end{cases} \quad (7 \text{ marks})$$

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(d) Calculate in eV the energy of the first excited state of a particle of mass 10^{-30} kg confined to one dimensional infinite square well potential of dimension 10^{-14} m. (4 marks)

QUESTION THREE

- a) i) Working from first principles, explaining the meaning of all symbols used and stating the assumptions made, derive an expression for discharge through an inclined venturimeter. [8 marks]
- ii) A venturimeter has an inlet diameter of 100mm and a throat diameter of 50mm with a flow rate of $15 \text{ dm}^3/\text{sec}$ of water. Calculate pressure head difference between the inlet and the throat in metres of water.
- iii) If the flow rate is doubles what would be the pressure head difference in meters of water? [6 marks]

[Total 20 marks]

QUESTION FOUR

- a) Derive an expression for the discharge of a liquid over a sharp edged V-notch of total angle 2θ degrees in terms of liquid depth H , angle θ degrees, the coefficient of discharge C_d and the acceleration due to gravity g . [10 marks]
 - b) A V-notch of total angle $91^\circ 24'$ discharges $410 \text{ dm}^3/\text{sec}$. Calculate the head above the bottom of the notch causing the discharge. [10 marks]
- Assume : $C_d = 0.6$
 $g = 9.81 \text{ m/sec}^2$



QUESTION FIVE

- a) i) Distinguish between turbines and pumps [2 marks]
- ii) Impulse turbines and reaction turbines [2 marks]
- b) Working from first principles, explaining the meaning all symbols used stating the assumptions made derive an expression for the hydraulic efficiency of a pelton wheel. [10 marks]
- c) A pelton wheel has a mean buduct speed of 10 m/se and the jet discharges $0.70 \text{ m}^3/\text{sec}$ over a head of 30.0 m. The deflection angle is 160° and C_v for nozzle is 0.98. Calculate the
 - i) Power developed [3 marks]
 - ii) Hydraulic efficacy [3 marks]

[Total 20 marks]